

Performance Modeling of a Cafeteria System

A discrete-event simulation study

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Outline

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- 2 Specification model
- 3 Simulator and verification
- 4 Transient analysis (Welch)
- 5 Experimental design and statistics
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Operational dilemma of a small cafeteria:

- Demand grows – should the manager hire a **waiter** or buy a **seat**?
- Intuition: “add a waiter when the queue grows”.
- Intuition is **not necessarily correct**: a customer occupies a seat throughout the entire stay, including consumption.

Three concrete questions answered in this study:

- 1 Which is the system's bottleneck, the waiters or the seats?
- 2 How sensitive are performance metrics to the arrival rate?
- 3 Given operating costs and a revenue model, what is the operational policy that maximises the hourly net benefit per demand level?

System description

Two customer classes (Bernoulli):

- CAFE: $p_c = 0.7$
- DESAYUNO: $p_d = 0.3$

Two resources:

- S **seats** (held for the whole stay)
- C **waiters** (held only during attention)

Two FCFS queues:

- Standing queue (unbounded)
- Seated-waiting queue (size $\leq S$)

Customer lifecycle:

```
ARRIVAL
↓ (if no seat)
standing queue
↓ seat allocated
seated-waiting
↓ waiter assigned
ATTENTION BEGINS
↓ END OF ATTENTION
consuming
↓ END OF CONSUMPTION
DEPARTURE (seat freed)
```

Why simulation?

In isolation: each resource would be an $M/M/c$ system.

In reality: the seat is also held **while the customer waits for a waiter**.

- The two resources are **coupled**.
- No closed-form expression exists for the joint distribution.

Methods used (all from course material):

- Algorithm 1.1 of Leemis & Park for the model-development workflow
- Algorithm 1.2 of Leemis & Park for the next-event simulation engine
- Multi-stream Lehmer generator for pseudo-random numbers
- Welch's method for warm-up identification
- Replication-deletion with t-Student CI for the statistical analysis

Specification model: parameters

Process	Distribution	Mean (min)
Inter-arrivals	Exponential	$1/\lambda$ (varied)
Class assignment	Bernoulli	$p_c = 0.7$ (coffee)
Attention – coffee	Exponential	$E[A_c] = 1.5$
Attention – breakfast	Exponential	$E[A_d] = 4$
Consumption – coffee	Exponential	$E[C_c] = 5$
Consumption – breakfast	Exponential	$E[C_d] = 12$

Why exponential service times?

- Memoryless property $\Rightarrow$ preserves Markov property in sub-models
- Allows analytical bounds for validation
- Standard reference in queueing theory

Derived means and analytical bounds

Marginal means averaged over the two classes:

$$E[A] = p_c E[A_c] + p_d E[A_d] = 0.7 \cdot 1.5 + 0.3 \cdot 4 = 2.25 \text{ min}$$

$$E[C] = p_c E[C_c] + p_d E[C_d] = 0.7 \cdot 5 + 0.3 \cdot 12 = 6.5 \text{ min}$$

$$E[T_{\text{seat}}] = E[A] + E[C] = 8.75 \text{ min}$$

Analytical utilization bounds (treating each resource in isolation):

$$\hat{\rho}_{\text{seat}}(C, S, \lambda) = \frac{\lambda E[T_{\text{seat}}]}{S}, \quad \hat{\rho}_{\text{waiter}}(C, S, \lambda) = \frac{\lambda E[A]}{C}$$

Key observation: $E[T_{\text{seat}}] \gg E[A] \Rightarrow$ the seat saturates first.

The seat is the predicted bottleneck.

Computational model

Implemented in ANSI C, following Algorithm 1.2 of Leemis & Park.

Next-event main loop:

- ➊ Initialize, schedule first arrival
- ➋ Find next event = $\min\{t_{\text{arrival}}, t_{\text{end-att}}[i], t_{\text{end-cons}}[j]\}$
- ➌ Advance clock to that time; update statistics; process event
- ➍ Repeat until $\text{clients_served} \geq 10\,000$

Multi-stream Lehmer RNG:

- $m = 2^{31} - 1$, $a = 48271$, 256 disjoint streams, jump multiplier 22 925
- One stream per stochastic process (decoupling):
 - Stream 0: inter-arrivals Stream 1: class
 - Streams 2-3: attention (coffee/breakfast)
 - Streams 4-5: consumption (coffee/breakfast)
- Variate generation by inverse method: $X = -m \ln(1 - U)$

Three distribution-free checks confirm the simulator:

(i) **Little's Law:** $L = \lambda W$ holds in every replication.

$$\text{Baseline: } \lambda W = \frac{1}{3} \cdot 16.06 = 5.36 = L \quad \checkmark$$

(ii) **Observed arrival rate** $= 1/E[\text{IAR}]$.

$$\text{Baseline: observed } X = 20.01 \text{ cust/h} = \lambda \quad \checkmark$$

(iii) **Class mix** converges to the input probabilities.

$$69.9\% \text{ coffee}/30.1\% \text{ breakfast} \approx 70/30 \quad \checkmark$$

Three independent checks $\Rightarrow$ the simulator is verified.

Welch's method – warm-up identification

Problem: the simulator starts empty $\Rightarrow$ early observations are biased.

Procedure (Leemis & Park, lect. 14):

- 1 Run $R = 30$ independent replications; take snapshots every $\Delta t = 60$ min.
- 2 Ensemble average across replications at each instant k :

$$\bar{Y}_k = \frac{1}{R} \sum_{r=1}^R Y_{r,k}$$

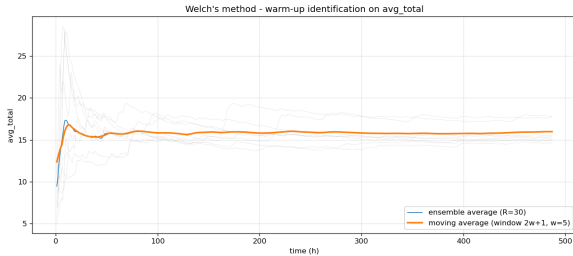
- 3 Smooth with a moving-average filter of window $2w + 1$ ($w = 5 \Rightarrow 11$ points):

$$\bar{Y}_k^{(w)} = \frac{1}{2w + 1} \sum_{s=-w}^w \bar{Y}_{k+s}$$

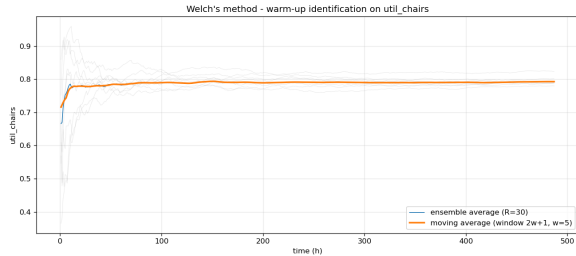
- 4 Visually identify T_0 = the instant when the smoothed curve stabilises.

Applied to 3 different indicators: cumulative W , instantaneous n in system, time-integrated seat utilization.

Welch's method – result



Cumulative mean response time



Time-integrated seat utilization

All three indicators stabilise by $t \approx 50$ h.

$T_0 = 60$ h with a 10 h safety margin

All subsequent statistics computed over $[T_0, T_{\text{end}}]$.

Statistical workflow and confidence intervals

Two-stage study:

- ① **Warm-up identification** (Welch, slide 11):
 $R = 30$ short replications on baseline $\Rightarrow T_0 = 60$ h
- ② **Steady-state estimation** (replication-deletion, deleting $[0, T_0]$):
 - **Baseline** $(C, S, \lambda) = (2, 4, 20)$ in detail: $n = 64$ replications
 - **Full factorial** $(C, S) \in \{2, 3\} \times \{4, 5\}$, $\lambda \in \{15, 20, 25, 30\}$:
 $n = 32$ replications $\times$ 16 scenarios = 512 runs

95% t-Student CI:

$$\bar{X} \pm t_{n-1, 0.025} \frac{s}{\sqrt{n}}, \quad t_{63} \approx 1.998 \text{ (baseline)}, \quad t_{31} \approx 2.040 \text{ (factorial)}$$

Why replication-deletion? Replications are **strictly independent** by construction (disjoint Lehmer streams) — no auto-correlation analysis needed.

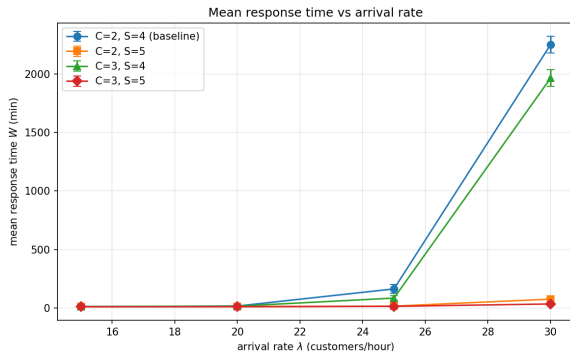
Baseline result: $(C, S, \lambda) = (2, 4, 20)$, $n = 64$ replications

Metric	Mean	St. dev.	95% CI half-width
W (min)	16.06	1.213	± 0.30 (1.9%)
Standing wait (min)	6.54	1.149	± 0.29 (4.4%)
Seated wait (min)	0.15	0.010	± 0.002 (1.6%)
Throughput (cust/h)	20.01	0.236	± 0.06 (0.3%)
Seat utilization	0.7935	0.014	± 0.0035 (0.4%)
Waiter utilization	0.3765	0.007	± 0.0017 (0.5%)
Mean number in system L	5.36	0.444	± 0.11 (2.1%)

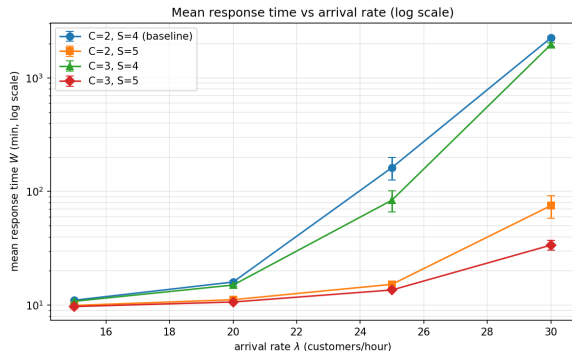
Sanity checks:

- Little: $\lambda W = \frac{1}{3} \cdot 16.06 = 5.36 = L$ ✓
- $X = 20.01 \approx \lambda$ (no losses, unbounded queue) ✓
- All half-widths $< 4.5\% \Rightarrow$ excellent statistical precision.

Response time across all scenarios



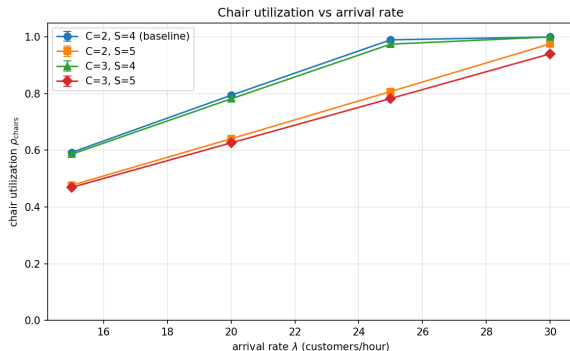
linear scale



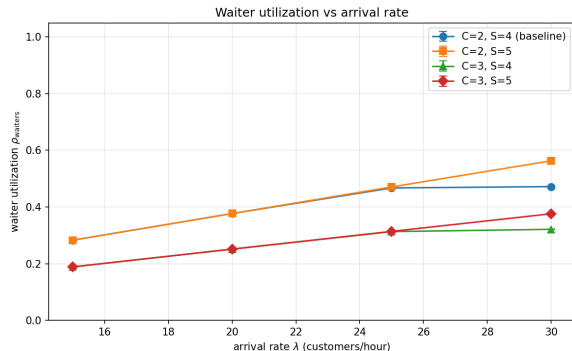
logarithmic scale

- Stable configurations: $W \in [10, 35]$ min across the whole range.
- $(2, 4)$ and $(3, 4)$ **saturate** at $\lambda = 30/h \Rightarrow W$ explodes.
- The log scale highlights the order-of-magnitude difference.

Bottleneck identification



Seat utilization



Waiter utilization (note same scale)

The seat is the bottleneck at every operating point:

- Baseline ($\lambda = 20$): $\rho_{\text{seat}} \approx 0.79$ vs $\rho_{\text{waiter}} \approx 0.38$
- Peak ($\lambda = 30$): seat **saturates at 1.0**, waiter stays below 0.56
- Confirms the analytical prediction $E[T_{\text{seat}}] \gg E[A]$.

Effect of adding one extra resource (at $\lambda = 25/h$)

Configuration	W (min)	Reduction vs baseline
$(C, S) = (2, 4)$ baseline	162.6	—
$(C, S) = (2, 5)$ +1 seat	15.3	−90%
$(C, S) = (3, 4)$ +1 waiter	84.0	−48%
$(C, S) = (3, 5)$ both	13.6	−92%

One extra seat $\rightarrow$ 90% lower W .

One extra waiter $\rightarrow$ only 48% lower.

The waiter upgrade is largely redundant: the bottleneck is so clearly the seat that $(3, 5)$ barely improves over $(2, 5)$ at this rate.

Validation against analytical predictions

Waiter side:

$\rho_{\text{waiter}}^{\text{sim}}$ agrees with $\hat{\rho}_{\text{waiter}}$ within ≤ 0.5 pp across all 16 scenarios.

$\Rightarrow$ Validates the waiter sub-model: a waiter behaves exactly as an isolated M/M/C.

Seat side:

$\rho_{\text{seat}}^{\text{sim}}$ is systematically 5–8 pp higher than $\hat{\rho}_{\text{seat}}$.

Reason: the bound ignores the time spent seated waiting for a waiter:

$$E[T_{\text{seat}}^{\text{real}}] = E[T_{\text{seat}}] + E[W_{\text{seated-waiting}}]$$

This coupling is precisely what defeats a closed-form solution and justifies the use of simulation.

Stability prediction: when $\hat{\rho}_{\text{seat}} > 1$, simulation shows saturation ($\rho^{\text{sim}} = 1.000$, W explodes) $\Rightarrow$ the bound is qualitatively correct.

Hourly net benefit

Benefit function (cost / revenue per hour):

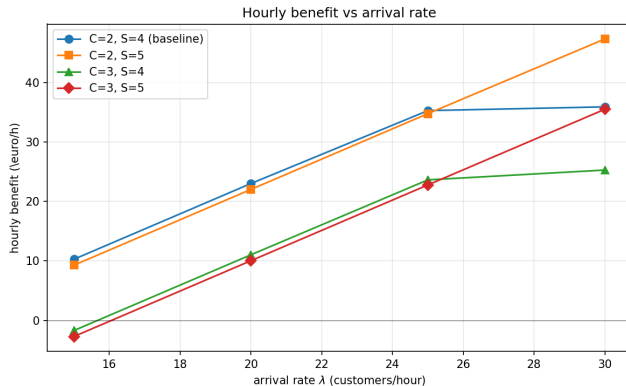
$$B(C, S, \lambda) = \underbrace{(p_c p_{\text{coffee}} + p_d p_{\text{breakfast}}) X(C, S, \lambda)}_{\text{revenue}} - \underbrace{(c_w C + c_s S)}_{\text{operating cost}}$$

Parameters:

- Prices: $p_{\text{coffee}} = 1.5 \text{ €}$, $p_{\text{breakfast}} = 5 \text{ €} \Rightarrow \bar{r} = 2.55 \text{ €/customer}$
- Costs: $c_w = 12 \text{ €/h}$, $c_s = 1 \text{ €/h}$

A seat is 12× cheaper than a waiter.

The chair cost amortises the capital expense (purchase, floor space, maintenance) over operating hours, allowing a like-for-like comparison with the waiter salary.



Recommended policy as a function of demand:

- $\lambda = 15/\text{h} \rightarrow (2, 4)$: cheapest, fine performance.
- $\lambda = 20/\text{h} \rightarrow (2, 4)$: baseline still wins.
- $\lambda = 25/\text{h} \rightarrow (2, 5)$: *one extra seat is enough.*
- $\lambda = 30/\text{h} \rightarrow (3, 5)$: both required, even so $W = 33.7$ min.

Rule: add a seat before adding a waiter.

Main findings:

- 1 The seat – not the waiter – is the bottleneck of the system at every operating point. The analytical bound $\hat{\rho}_{\text{seat}} = \lambda E[T_{\text{seat}}]/S$ correctly predicts both the magnitude and the stability boundary.
- 2 Adding a seat is operationally and economically far superior to adding a waiter: 12× cheaper and directly addresses the bottleneck.
- 3 The optimal configuration is demand-dependent: (2, 4) at low to normal demand, (2, 5) at high demand, (3, 5) only at peak.

Limitations and possible extensions:

- Service times assumed exponential (standard but not always accurate).
- Customer balking and reneging not modelled.
- Extensions: heterogeneous waiters, time-varying λ (peak hours), take-away service not needing a seat.

Thank you.

Questions?